

AFL Assistant (Executive Report)

1. Product Goal

The AFL Assistant is a domain-locked conversational system designed to answer AFL questions, perform grounded AFL predictions, maintain multi-turn context, and refuse requests outside its AFL scope. The system combines LangGraph orchestration, structured AFL data/tools, prediction models, guardrails, and a FastAPI interface suitable for local demonstration or further deployment.

2. Architecture & Key Capabilities

Layer	Purpose
Input / FastAPI	Accepts message + conversation_id and returns response, intent, prediction metadata and latency.
LangGraph routing	Classifies the request and routes factual, prediction, and out-of-scope queries.
AFL factual tools	Provides rule/stat/team/player answers grounded in the AFL domain.
Prediction tools	Supports match-winner and top-player predictions with validation and safe tool invocation.
Guardrails	Reject prompt-injection attempts, off-topic requests, unsupported prediction types, and invalid matchups.
Monitoring	Records query, intent, tools called, latency, token usage and errors in structured logs.

3. Evaluation Results

A combined 25-case evaluation was supplied covering factual accuracy, guardrails, multi-turn coherence, and prediction sanity. All 25 cases passed, giving an overall pass rate of 100%.

Category	Cases	Passed	Pass rate
Factual Q&A	7	7	100%
Guardrails / injection	6	6	100%
Multi-turn coherence	6	6	100%
Prediction sanity	6	6	100%
Overall	25	25	100%

Evaluation observations

- **Factual:** 7/7 passed. Basic AFL rules and terminology were answered correctly.
- **Guardrails:** 6/6 passed. Prompt-injection attempts such as requests to reveal the system prompt or disable AFL restrictions were refused.
- **Multi-turn:** 6/6 passed. Follow-up questions such as “What about teams?” remained within the AFL context.
- **Prediction:** 6/6 passed. Match-winner and top-player predictions returned successfully; unsupported exact-score requests and same-team matchups were rejected.
- **Weakest category:** No category was weaker by pass rate; all categories achieved 100%. The next concrete improvement should therefore target prediction grounding/data freshness, especially replacing unknown match prediction data cut-off metadata with an explicit as-of date.

4. Prediction & Model Safety Findings

The evaluation confirms that the prediction layer is operational and applies a consistent disclaimer: **“Prediction, not a certainty.”** The match-winner tool returned probabilities such as 51.7% and 57.7%, while the top-player tool returned ranked predicted fantasy points. Unsupported exact-score and winning-margin requests were explicitly rejected.

A key limitation is visible in the match-winner response: the grounding text reports the available data date as **unknown**. This weakens transparency even though the prediction itself is returned. The top-player prediction has stronger metadata, including eligibility seasons and a data-through date.

Recommended improvement: expose a reliable model/data `as\_of\_date` from the prediction layer for every match-winner response, and use it consistently in both the natural-language response and prediction metadata.

The supplied test set does not contain a historical accuracy comparison against a public/ladder-position baseline, so no defensible benchmark superiority claim is made in this report. For production evaluation, compare the model against a simple baseline such as selecting the higher ladder-ranked team, using identical historical match dates and an out-of-sample test period.

5. Known Limitations

- Prediction accuracy is not guaranteed; probabilities are model outputs rather than certainties.
- Available data can become stale as new rounds, transfers, injuries and team-list changes occur.
- The current prediction scope is intentionally limited to match winners and top players; exact scores and margins are unsupported.
- The supplied evaluation is a functional/behavioral test suite and does not establish statistical model superiority.
- One measured factual test had high latency (~9.1 seconds), showing that latency monitoring should remain part of production hardening.

6. Monitoring & Maintenance Checklist

Metric	Suggested alert / action	Cadence
API latency	Investigate sustained p95 > 2s; urgent review for repeated multi-second spikes.	Continuous / weekly review
Tool error rate	Alert if >5% over a rolling window.	Continuous
Off-topic leak rate	Target 0%; any successful leak triggers guardrail test/update.	Every release
Prompt-injection pass rate	Target 100% on regression suite.	Every release
Prediction accuracy	Track calibration, accuracy and Brier score on newly completed matches.	After each round
Data freshness	Refresh feature tables when new official match/player data arrives.	Weekly / each round
Retraining	Retrain when new rounds materially expand training data or drift is detected.	1. Weekly candidate run

7. Weekly Refresh Loop

1) Ingest newly completed AFL match/player results. 2) Validate schema, dates, duplicates and missing values. 3) Append/update the feature tables. 4) Recalculate rolling/player-form features without using future information. 5) Run the regression/evaluation suite and compare the prediction model with the baseline. 6) Retrain only when evaluation is acceptable and the new model improves or maintains agreed metrics. 7) Version the model and record the training-data cut-off. 8) Deploy the new version and monitor post-release drift.

8. Recommended Next Steps

- Add explicit model/data cut-off metadata to match-winner predictions.
- Automate the 25-case regression suite in CI so guardrail and conversational regressions block releases.
- Add a ladder-position baseline and report out-of-sample model-vs-baseline metrics.

- Add request/tool timeouts and latency-based monitoring around prediction tools.
- Build the optional Streamlit/simple HTML UI for a stakeholder-friendly live demo.

Conclusion: The current AFL Assistant demonstrates a complete capstone architecture with a working API, prediction tools, domain guardrails, multi-turn behavior, and structured evaluation. The supplied evaluation achieved 100% across all 25 functional cases. Before production use, the highest-value hardening work is improving prediction data transparency, adding a public-baseline comparison, automating regression testing, and continuously monitoring latency, errors, guardrail leakage and model drift.